Joshua Brown
Web and Cloud
Labs 9

9.1g

client secret: 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
client id: 780966768163-ek6nhkn6qghqcf484agsp0jdg2687tkn.apps.googleusercontent.com

```
None
gcloud run services update gcp-oauth-gb \
  --update-env-vars
REDIRECT_CALLBACK=https://gcp-oauth-gb-780966768163.us-west1.run.app
/callback \
  --update-env-vars
CLIENT_ID=780966768163-ek6nhkn6qghqcf484agsp0jdg2687tkn.apps.googleu
sercontent.com \
  --update-env-vars
CLIENT_SECRET=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 \
  --region=us-west1
```

Take a screenshot of the Headers that includes the URL and the returned HTTP status code for the first two requests for your lab notebook.

gcp-oauth-gb-780966768163.us-west1.run.app

Sign Guestbook

J Joshua Brown <jjb24@pdx.edu>

Console | Sources | Network | Timelines | Storage | Graphics | Layers | Audit

Disable Caches | Import | Export

Preview | Headers | Cookies | Sizes | Timing | Security

Summary

URL: https://gcp-oauth-gb-780966768163.us-west1.run.app/callback?state=Y5HhD5kerOdYwgnpjv373NTZUqgYhN&code=4%2F0Ab32j92wtKXMjf0GC-bscGbJa0hzvYQhw-J7LKQpJXNQDunlnKjzwznxBMlwU338ZPbnXw&scope=email+profile+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.profile+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email+openid&authuser=0&hd=pdx.edu&prompt=consent

https://gcp-oauth-gb-780966768163.us-west1.run.app/sign

Status: 200
Source: Network
Address: 2600:1900:4241:200:::443

Request

GET /callback
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.9
Priority: u=0, i
Referer: https://accounts.google.com/
Sec-Fetch-Dest: document
Sec-Fetch-Mode: navigate
Sec-Fetch-Site: cross-site
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/26.1 Safari/605.1.15

Redirect Response

gcp-oauth-gb-780966768163.us-west1.run.app

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Console | Sources | Network | Timelines | Storage | Graphics | Layers | Audit

Disable Caches | Import | Export

Preview | Headers | Cookies | Sizes | Timing | Security

Summary

URL: https://gcp-oauth-gb-780966768163.us-west1.run.app/static/style.css
Status: 304
Source: Memory Cache
Address: 2600:1900:4241:200:::443
Initiator: sign:4

Request

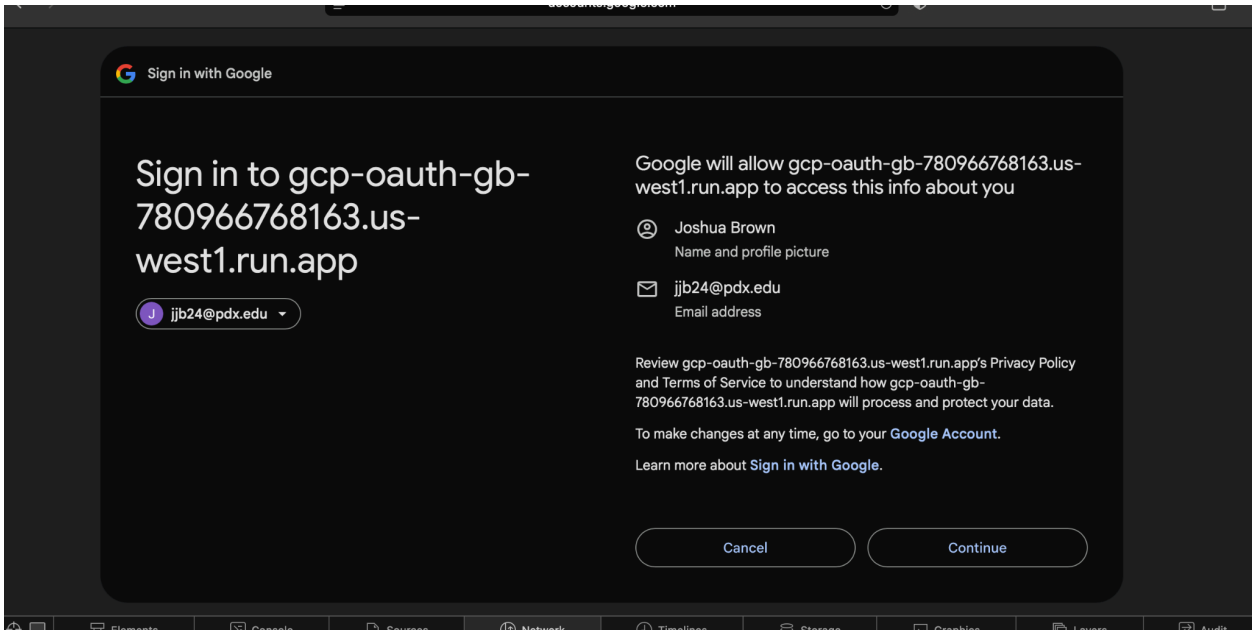
:method: GET
:scheme: https
:authority: gcp-oauth-gb-780966768163.us-west1.run.app
:path: /static/style.css
Accept: text/css,*/*;q=0.1
Accept-Encoding: gzip, deflate, br
Accept-Language: en-US,en;q=0.9
Cookie: session=.eJy9VMmSozgU_Bef21UgjG3qBma3EUUZLNDEhAMDBrEbMFtH_3vjjtkOc5jTnBRKPb18kRkvv68q_9kl17bzu2j1sfJYNRHZLGrM0Bvisk57ZsdAGzuP2Evg6tsf5V2VReXq4_vKD4Kobf-8ryYfcG8-xdtkewS97plsJbF6JiHr2PHYHoYplp_r7bOjoD6VDu3aufGsVCZdl02SHVFLNbPXnEzY5iatSnYsTum5f9yrKy3kJ6ytLd8kzQQRks3kpmZGud8KrGKJpbd92EFfS4XrUlCuhYjpY3hXWsBYgFKHB0m3lnMxNHx-6Kltf6pR4NSjtr6ZaPgcggztD1Zs7Qp3gHDgN4RlyeAfYu_lQ86Z_7LOsqUqLjClMzMKOgzsqA6i1mv660gnCU8BaruoE401aaL26nerD3q33bDshttyf-NkEYlhuQUh4V-iRZOe3JSAmEQ_Q7NGQ6K1WHHV-KBtiTSbPTuboZ2kHvBYE3mjV2gDFAXizd5kpHlGbYs1gDPiAmemImeeLZHTQacll3_1PF5oi9ytt4WmDhjjBSUegH2lWCqY2PlGuPKmOE8PcJ1GBuK59aClFYFzPJj2Uji7G8P25hO6NLBgKx9sQlDCKkCwgGJMeSicA1F4YCWYYBozodl9T4U8BKrxxIDrMRjzQIV5UMD-Vn7lt9L6B9cl-7-4AuZCXlyGzVPQjifDligo8q9zgLMDDNGbXnW-Yr00GrAaPzH6ol8YRnTio-GFP_xCm6Ak_Mv7eA-Ly-QjXGO09Cl_phA5i5Uy5YlkCUD8q6-XXg4GYwCkYtbJ4yG86LqftazF6MwvfiRTy5zjosUypzMaczAsduaRutiZSgxMLfplOZwF7i1hLgXvg0fT7neUCsx9mzz7Jmnk4rrTWZUlFydqrOSg1VyRXz_raNj7RJzvlESVd0dNYbMWWpbgA9NkcQUo_nD1yzCiz3Ys7QglhHhcc_pGjc46T99DecCpwxSfhxvf0vv9sohT-6g5Q7yqSpaQGxn5g_U8h26oYk8pwHi_NF69JcE0fW7cbOtunoXuhErW36qcqzOA08_brGf-5lQplTmze7r1dhenk-FVbE_urGiP-nrVgXDjtPTxhQ7dqSp9lEi9l7UIuxlhWjzypx7dy-qYlrFsMZRUVwjAPevPlbfJU9492RJXyFkzH1rossDmH51OEE5SHsxzDE4P6iodEW2mNe6v43lYNrkNqnoJxN9WSdfV7cf7-zAMb3FVxXnk16R9C6ri_RWF7882akh5r96iwif58vM_f6ib6k7y6BWqdVSScPX7t9WvTLh204t7JUR-EzWrHz9-Ahz47rQ.aSzqyg.ckScUMfkAyOD2WV7zu-lnYJr7_4
If-Modified-Since: Mon, 01 Dec 2025 00:24:05 GMT
If-None-Match: "1764548645.0-566-1438189567"
Priority: u=1, i
Referer: https://gcp-oauth-gb-780966768163.us-west1.run.app/sign

2 | 3 | 2.19 KB | 3.27 KB | 1 | 572ms

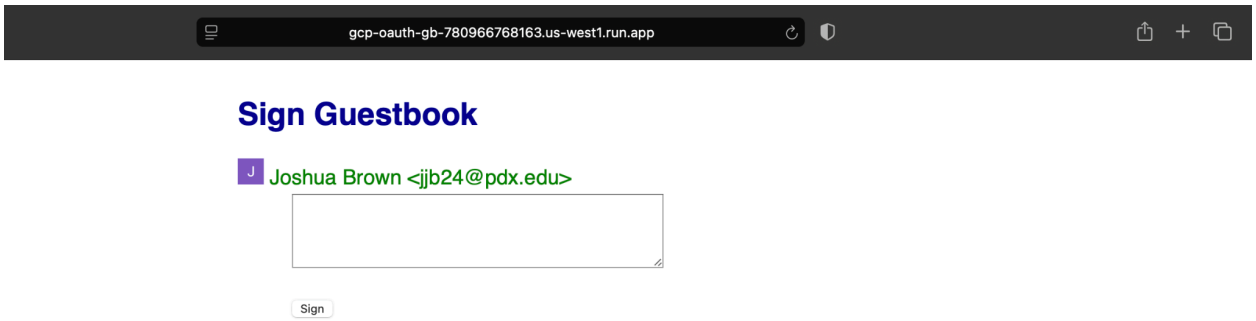
Based on the description of the source code, what lines of code in our application are responsible for the second request shown?

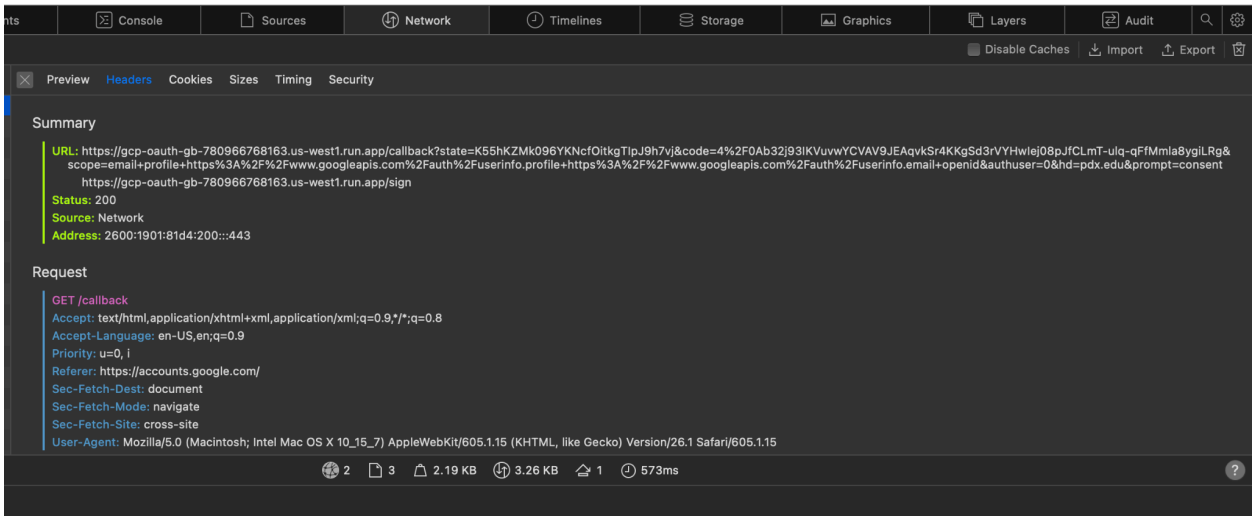
```
from oauth_config import client_id, authorization_base_url, redirect_callback
...
# If OAuth token not present, start OAuth flow
google = OAuth2Session(
          client_id,
          redirect_uri = redirect_callback,
          scope = 'https://www.googleapis.com/auth/userinfo.email ' +
                  'https://www.googleapis.com/auth/userinfo.profile'
        )
```

Take a screenshot of the permissions you (as a user) are granting the Guestbook access to.



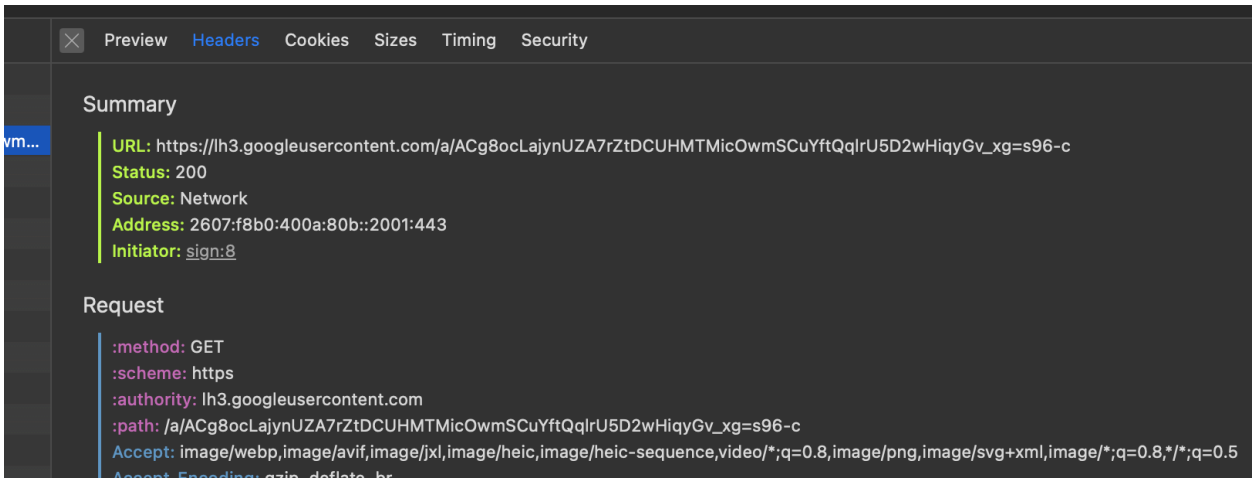
Take a screenshot of the Headers that includes the entire Callback URL and its returned HTTP status code. What is the URI for the Location that the User sent to by the Callback?



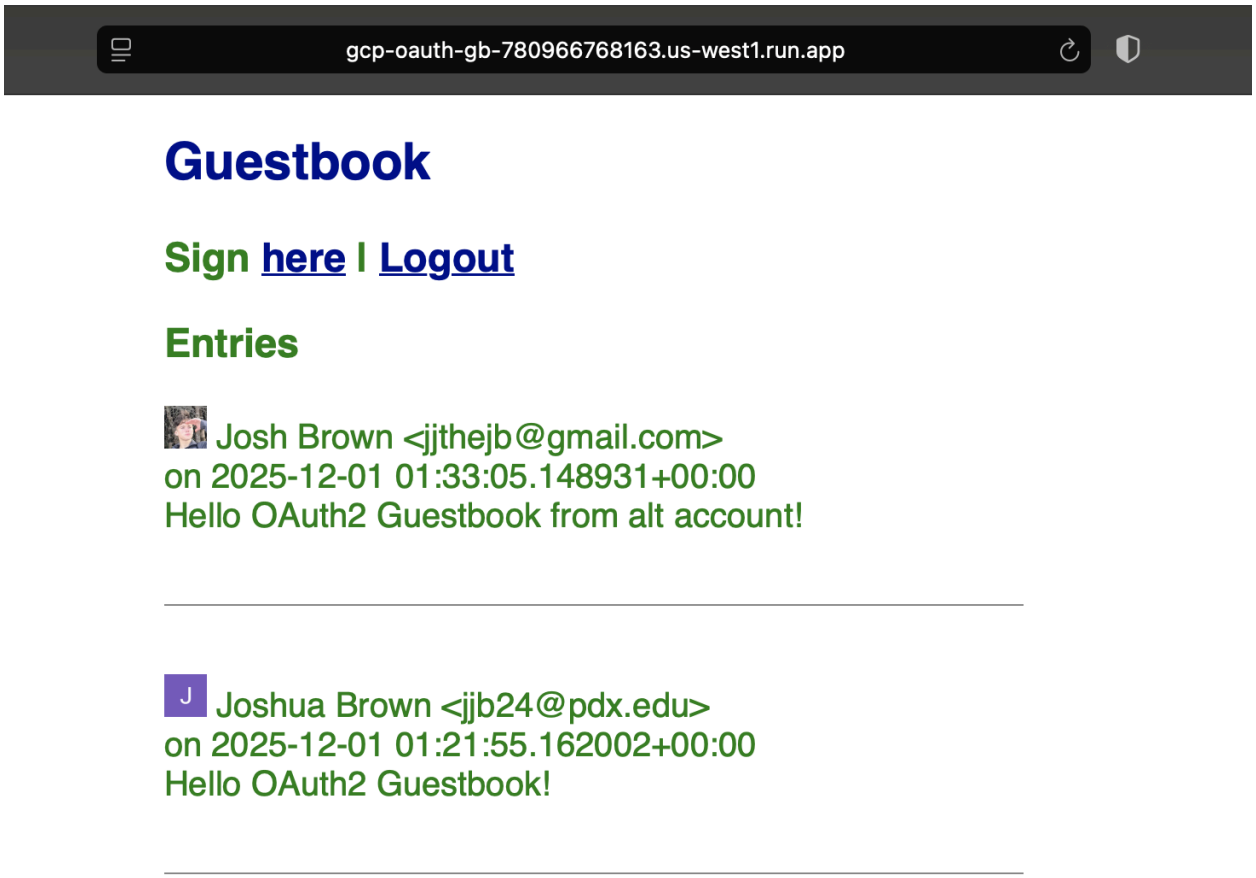


https://gcp-oauth-gb-780966768163.us-west1.run.app/sign

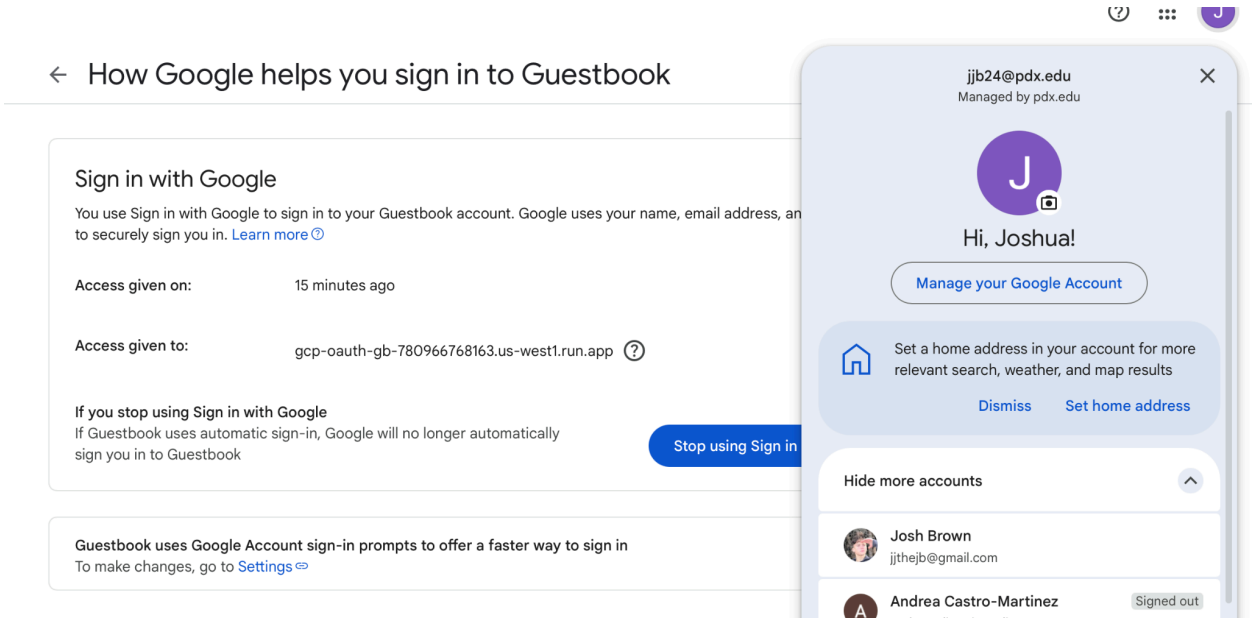
Find the request within Developer Tools that fetches the embedded image and take a screenshot of its URL.



Take a screenshot showing multiple authenticated accounts have been able to sign the Guestbook.



Take a screenshot of the expanded information that includes your OdinId for your lab notebook.



9.2g

What other domains are given access to this Firebase project by default?

Authorized domain	Type
localhost	Default
fir-jjb24.firebaseapp.com	Default
fir-jjb24.web.app	Default
cloudshell.dev	Custom

Take a screenshot of the first 10 lines of the produced file.

```
==> firebase-messaging-sw.js <==
/*
 * ATTENTION: An "eval-source-map" devtool has been used.
 * This devtool is neither made for production nor for readable output files.
 * It uses "eval()" calls to create a separate source file with attached SourceMaps
in the browser devtools.
 * If you are trying to read the output file, select a different devtool (https://we
bpack.js.org/configuration/devtool/)
 * or disable the default devtool with "devtool: false".
 * If you are looking for production-ready output files, see mode: "production" (htt
ps://webpack.js.org/configuration/mode/).
 */
/******/ (() => { // webpackBootstrap
/******/        var __webpack_modules__ = ({
jjb24@cloudshell:~/codelab-friendlychat-web/web-start/public (firebase-jjb24)$
```

config code:

```
const firebaseConfig = {

  apiKey: "AIzaSyA_Lkp1osiPkfUeo0akzrUj3A6awRDJesQ",
  authDomain: "fir-jjb24.firebaseapp.com",
  projectId: "firebase-jjb24",
  storageBucket: "firebase-jjb24.firebasestorage.app",
  messagingSenderId: "840893645064",
  appId: "1:840893645064:web:1efaf8f7326b9239f1ecbd"
};
```

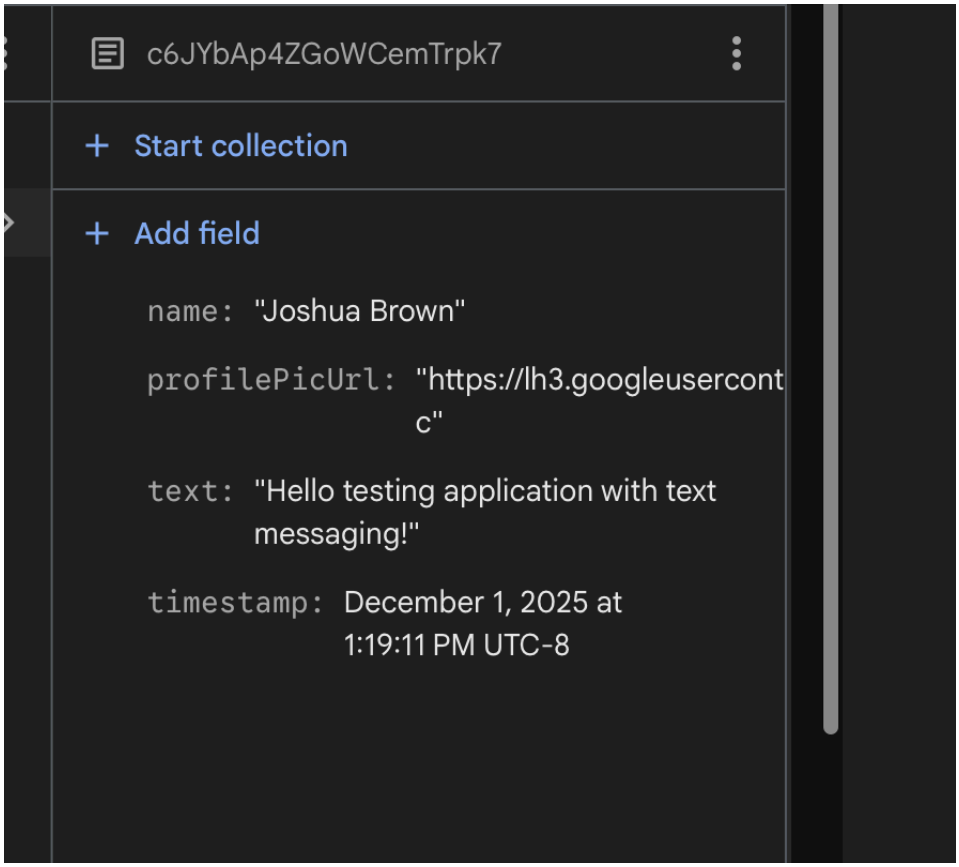
What missing functions deal with user authentication?
signIn(), signOutUser(), initFirebaseAuth()

What missing functions deal with sending and receiving messages?
SaveMessage(), LoadMessages(), saveImageMessage() saveMessagingDeviceToken()

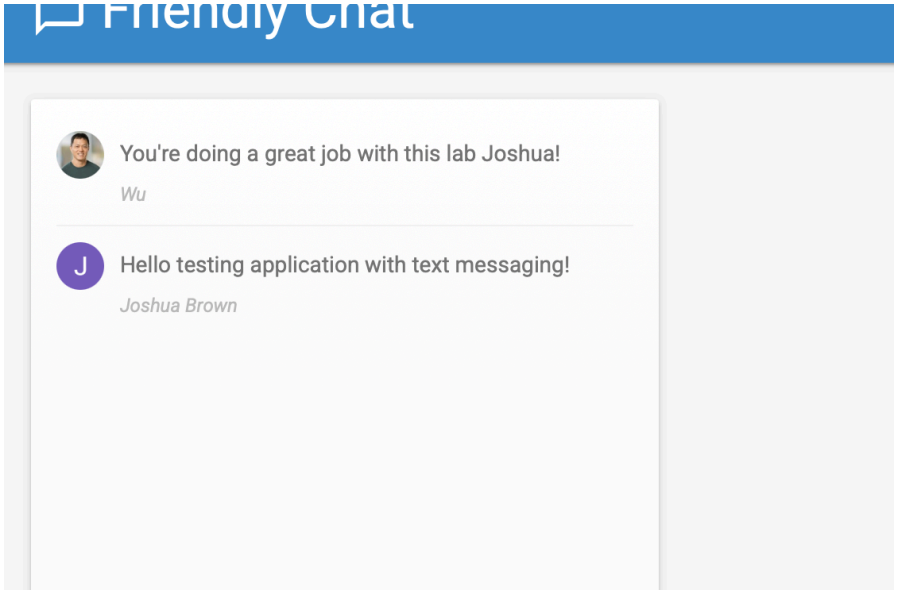
What are the names of the elements that are hidden when the user is signed out?
userNameElement, userPicElement, signOutButtonElement

What is the name of the element that is not hidden when the user is signed out?
signInButtonElement

Include a screenshot of the message and its fields in the database for your lab notebook



Include a screenshot of the application with its two messages for your lab notebook

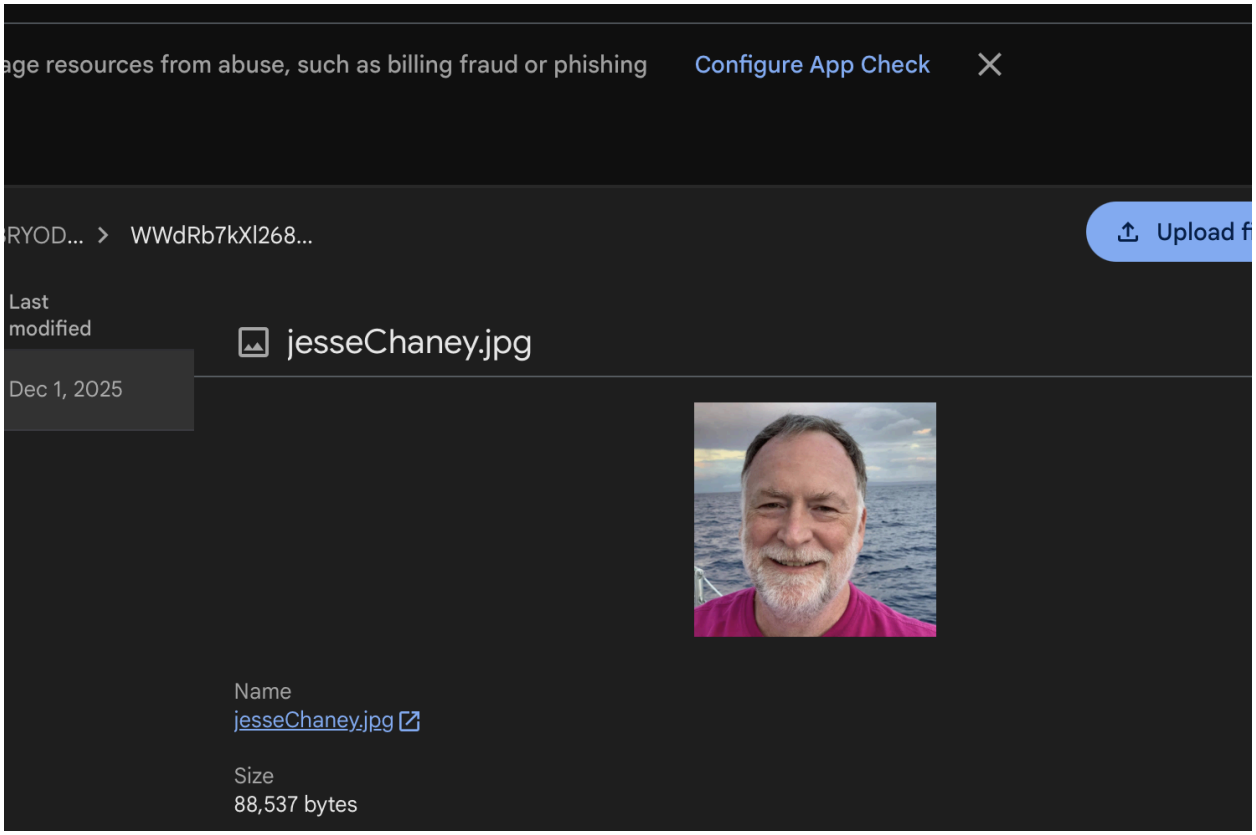


What is the URL of the image that is first shown in the UI as the message is loading?
https://www.google.com/images/spin-32.gif?a

How do the fields in an image document differ from that of the text document?
There is an additional imageUrl and storageUri, and no message field

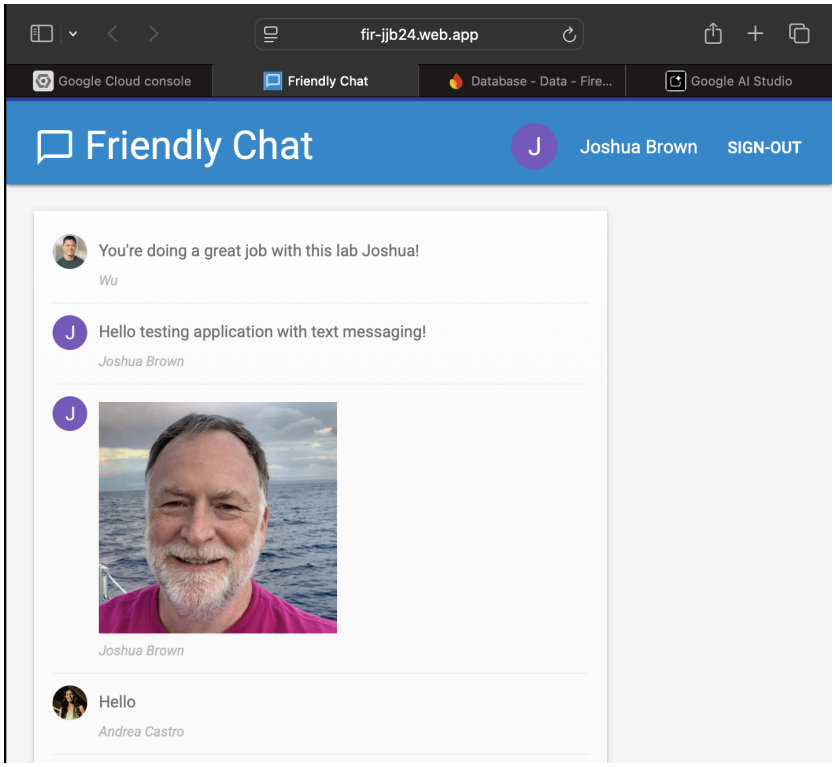
What URL and storage location can the image be found at?
Storage location: 4JcoQojBRYODYSdlTKNizShin7G3
imageUrl:
"https://firebasestorage.googleapis.com/v0/b/firebase-jjb24.firebasestorage.app/o/4JcoQojBRYODYSdlTKNizShin7G3%2FWWdRb7kXl268bhTCMxdm%2FjesseChaney.jpg?alt=media&token=68af77a4-f78a-4289-983a-6a1d618a6880"

Take a screenshot of the image in the storage bucket for your lab notebook.



What directory is the application going to be served from?
public

Take a screenshot of the message including the URL for your lab notebook.



9.3g

Take a screenshot of the table's details that includes the number of rows in the table.

Table info

Edit

Table ID	cloud-brown-jjb24.yob.yob_native_table
Created	Dec 1, 2025, 1:52:22 PM UTC-8
Last modified	Dec 1, 2025, 1:52:22 PM UTC-8
Table expiration	NEVER
Data location	us-west1
Default collation	
Default rounding mode	ROUNDING_MODE_UNSPECIFIED
Case insensitive	false
Description	
Labels	
Primary key(s)	
Tags	

Storage info

Number of rows	33,044
Total logical bytes	618.78 KB
Active logical bytes	618.78 KB

Screenshot the query results and include it in your lab notebook

Job information | Results | Visualization | JSON | Execu

Row	name	count
1	Emma	20799
2	Olivia	19674
3	Sophia	18490
4	Isabella	16950
5	Ava	15586
6	Mia	13442
7	Emily	12562
8	Abigail	11985
9	Madison	10247
10	Charlotte	10048
11	Harper	9564
12	Sofia	9542
13	Avery	9517
14	Elizabeth	9492
15	Amelia	8727

Screenshot your results and include it in your lab notebook

```
jjb24@cloudshell:~ (cloud-brow
wn-jjb24.yob.yob_native_table]
+---------+-------+
|  name   | count |
+---------+-------+
| Aari    |     5 |
| Aaliyah |     5 |
| Aadian  |     5 |
| Aaroh   |     5 |
| Aarit   |     5 |
| Aadiv   |     5 |
| Aadhi   |     5 |
| Aarohan |     5 |
| Aariyan |     5 |
| Aamer   |     5 |
+---------+-------+
jjb24@cloudshell:~ (cloud-brow
```

Screenshot your results and include it in your lab notebook

```
WHERE gender= M' ORDER BY c
+-----------+-------+
|   name    | count |
+-----------+-------+
| Noah      | 19144 |
| Liam      | 18342 |
| Mason     | 17092 |
| Jacob     | 16712 |
| William   | 16687 |
| Ethan     | 15619 |
| Michael   | 15323 |
| Alexander | 15293 |
| James     | 14301 |
| Daniel    | 13829 |
+-----------+-------+
cloud-brown-jjb24>
```

Screenshot your results and include it in your lab notebook

```
cloud-brown-jjb24> SELECT name, c
WHERE name='Josh';
+------+-------+
| name | count |
+------+-------+
| Josh |   163 |
+------+-------+
cloud-brown-jjb24> SELECT name, c
WHERE name='Joshua';
+--------+-------+
|  name  | count |
+--------+-------+
| Joshua |    15 |
| Joshua | 10764 |
+--------+-------+
cloud-brown-jjb24>
```

9 - Screenshot the query results and include it in your lab notebook

Job information | Results | Visualization | JSON | Execu

Row	name	count
1	Aarshi	5
2	Aaniylah	5
3	Aaryah	5
4	Aashirya	5
5	Aalimah	5
6	Aarielle	5
7	Aarabella	5
8	Aayra	5

Results per page: 50 1 – 20 of 20

9.4g

How much less data does this query process compared to the size of the table?

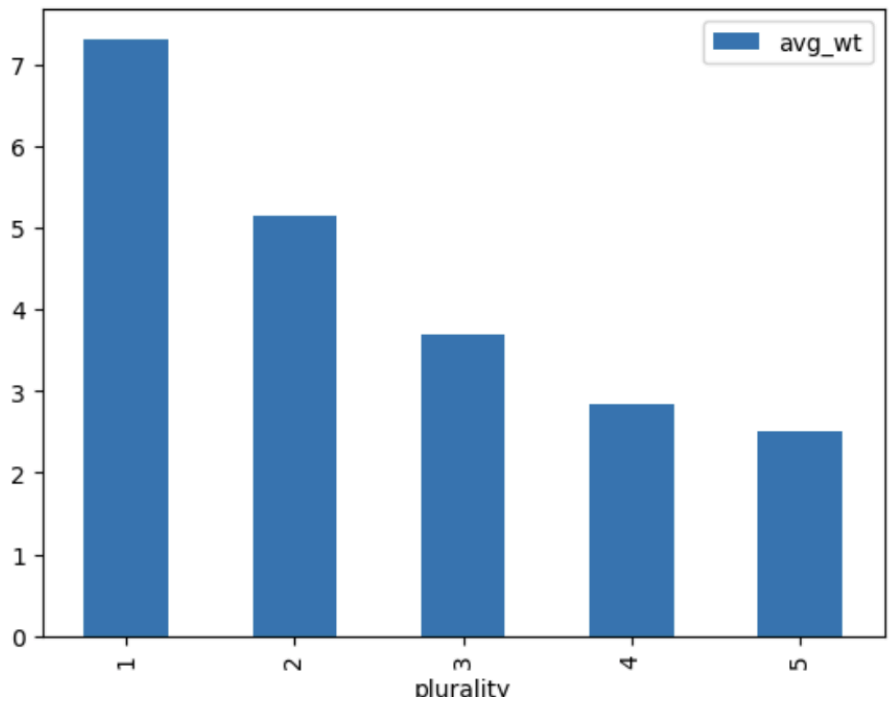
This query is almost 20gb less than the size of the table

How many twins were born during this time range?
375,362

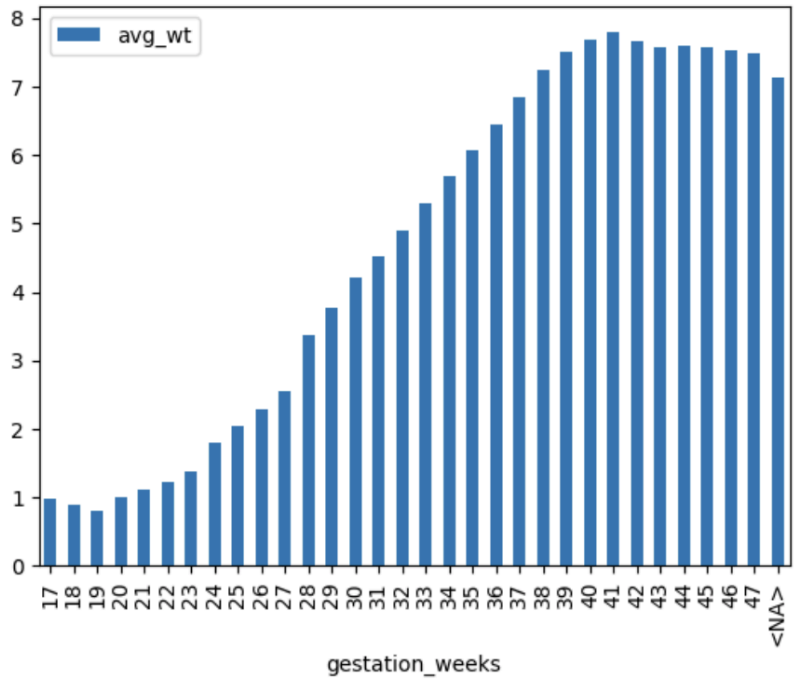
How much lighter on average are they compared to single babies?
on average, twins are about 2.18 pounds lighter

Show the plots generated for the two most important features for your lab notebook

```
<Axes: xlabel='plurality'>
```



```
0]: <Axes: xlabel='gestation_weeks'>
```



What day saw the largest spike in trips to grocery and pharmacy stores?
6/29/24

On the day the stay-at-home order took effect (3/23/2020), what was the total impact on workplace trips?
-11

Which three airports were impacted the most in April 2020 (the month when lockdowns became widespread)?
Detroit, McCarran, San Francisco

Run the query again using the month of August 2020. Which three airports were impacted the most?
McCarran, Detroit, San Francisco

What table and columns identify the place name, the starting date, and the number of excess deaths from COVID-19?
table: excess_deaths
columns: place_name, start_data, total_deaths, expected_deaths

What table and columns identify the date, county, and deaths from COVID-19?
table: us_states
columns: date, state_name, confirmed_cases

What table and columns identify the date, state, and confirmed cases of COVID-19?
us_states
columns: date, state_name, confirmed_cases

What table and columns identify a county code and the percentage of its residents that report they always wear masks?
table: mask_use_by_county
columns: county_fips_code, ALWAYS

Show a screenshot of the plot and the code used to generate it for your lab notebook

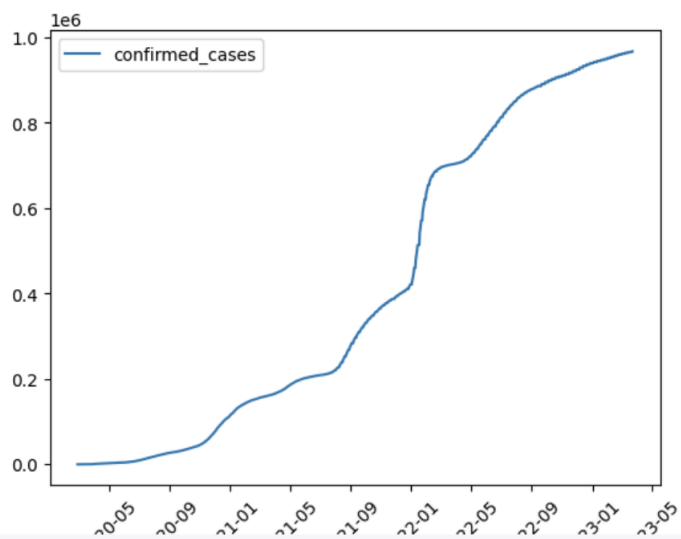
[14]:
```
%%bigquery df
SELECT date, confirmed_cases
FROM `bigquery-public-data.covid19_nyt.us_states`
WHERE state_name = 'Oregon'
ORDER BY date ASC
```

Job ID d0423d0d-c71b-4bd8-abcb-86903a61b6b9 successfully executed: 100%

Downloading: 100%

[15]:
```
df.plot(x='date', y='confirmed_cases', kind='line', rot
```

[15]:
```
<Axes: xlabel='date'>
```



From within your Jupyter notebook, run the query and write code that shows the first 10 states that reached 1000 deaths from COVID-19. Take a screenshot for your lab notebook.

[16]:
```
%%bigquery df_deaths_by_state
SELECT state_name, MIN(date) as date_of_1000
FROM `bigquery-public-data.covid19_nyt.us_states`
WHERE deaths > 1000
GROUP BY state_name
ORDER BY date_of_1000 ASC
```

Job ID d809b08e-90ec-45bc-885c-29b4a79237e5 successfully executed: 100%

Downloading: 100%

[17]:
```
df_deaths_by_state.head(10)
```

[17]:

	state_name	date_of_1000
0	New York	2020-03-29
1	New Jersey	2020-04-06
2	Michigan	2020-04-09
3	Louisiana	2020-04-14
4	Massachusetts	2020-04-15
5	Illinois	2020-04-16
6	California	2020-04-17
7	Connecticut	2020-04-17
8	Pennsylvania	2020-04-17
9	Florida	2020-04-24

Take a screenshot for your lab notebook of the Top 5 counties and the states they are located in.

[18]:
```
%%bigquery df_mask_usage
SELECT DISTINCT ct.county, ct.state_name, mu.always
FROM `bigquery-public-data.covid19_nyt.mask_use_by_cour
LEFT JOIN `bigquery-public-data.covid19_nyt.us_counties
ON mu.county_fips_code = ct.county_fips_code
WHERE ct.county IS NOT NULL AND ct.state_name IS NOT NU
ORDER BY mu.always DESC
```

Job ID 49cf1352-f97b-4fd1-9998-

7c2aed940a54 successfully executed: 100%

Downloading: 100%

[19]:
```
df_mask_usage.head(5)
```

[19]:

	county	state_name	always
0	Inyo	California	0.889
1	Yates	New York	0.884
2	Hudspeth	Texas	0.880
3	Mono	California	0.880
4	El Paso	Texas	0.877

Plot the results and take a screenshot for your lab notebook.

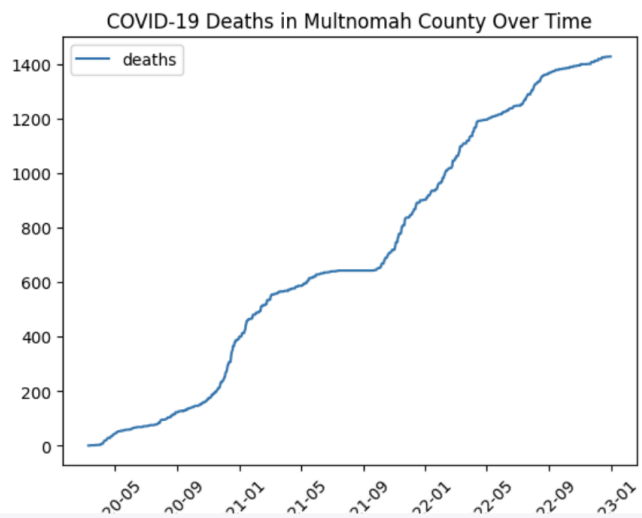
[20]:
```
%%bigquery df_multnomah
SELECT date, deaths
FROM `bigquery-public-data.covid19_nyt.us_counties`
WHERE county = 'Multnomah' AND state_name = 'Oregon'
ORDER BY date ASC
```

Job ID aaecce39-b023-4bb8-b468-557a11f47003 successfully executed: 100%

Downloading: 100%

[21]:
```
COVID-19 Deaths in Multnomah County Over Time', rot=45)
```

[21]:
```
<Axes: title={'center': 'COVID-19 Deaths in Multnomah
County Over Time'}, xlabel='date'>
```

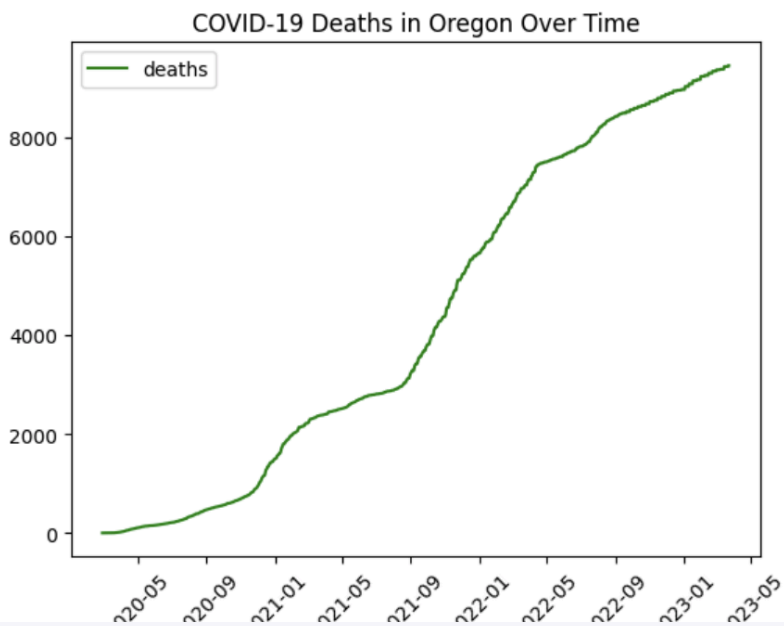


Plot the results and take a screenshot for your lab notebook.

[22]:
```
%%bigquery df_oregon
SELECT date, deaths
FROM `bigquery-public-data.covid19_nyt.us_states`
WHERE state_name = 'Oregon'
ORDER BY date ASC
```

Job ID 763efc5b-0f06-4c52-8167-4022546aa875 successfully executed: 100%

Downloading: 100%

[23]:
```
-19 Deaths in Oregon Over Time', color='green', rot=45)
```

[23]:
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<Axes: title={'center': 'COVID-19 Deaths in Oregon Over Time'}, xlabel='date'>
```